Linear Models I

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Linear Models

- Linear regression model: a very simple approach for supervised learning and a useful tool for predicting a quantitative response.
- Linear regression has been around for a long time and is still very useful in statistical learning and machine learning.
- Linear regression model can serve as a good jumping-off point for newer approaches.

A Motivating Example

Plot below displays sales (in thousands of units) for a particular product as a function of advertising budgets (in thousands of dollars) for TV, radio, and newspaper media.

Uses of Linear Models

Here are a few important questions that a linear regression model can answer

- Is there a relationship between advertising budget and sales?
- Does the data provide any evidence of an association between advertising expenditure and sales. If the evidence is weak, then one might argue that no money should be spent on advertising!
- How strong is the relationship between advertising budget and sales? If there is a relationship between advertising and sales, we want to know the strength of this relationship.

Uses of Linear Models

- Which media contribute to sales? We need to separate out the individual effects of each of three media when we have spent money on all three media.
- How accurately can we estimate the effect of each medium on sales? For every dollar spent on advertising in a particular medium, by what amount will sales increase? How accurately can we predict this amount of increase?

Uses of Linear Models, continued

- How accurately can we predict future sales? For any given level of television, radio, or newspaper advertising, what is our prediction for sales, and what is the accuracy of this prediction?
- Is there synergy among the advertising media? Perhaps spending \$10,000 on television advertising and \$10,000 on radio advertising results in more sales than allocating \$20,000 to either television or radio individually. In marketing, this is known as a synergy effect, while in statistics it is called an interaction effect.

Uses of Linear Models, continued

- Is the relationship linear? If there is approximately a straight-line relationship between advertising expenditure in the various media and sales, then linear regression is an appropriate tool. If not, then it may still be possible to trans- form the predictor or the response so that linear regression can be used.

Simple Linear Models

- Simple linear regression: predicting a quantitative response Y with a single predictor variable X by assuming approximately a linear relationship between X and Y .
- Mathematically, we can write this linear relationship as

$$Y \approx \beta_0 + \beta_1 X,$$

where $\approx$ is read as "is approximately modeled as".

Simple Linear Models

- In the equation above, β_0 and β_1 are two unknown constants that represent the intercept and slope terms in the linear model, known as the model coefficients or parameters.
- For example, X may represent TV advertising and Y may represent sales. Then we can regress sales onto TV by fitting the model:

$$\text{sales} \approx \beta_0 + \beta_1 \text{ TV} .$$

Notation

- $$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2, S_{yy} = \sum_{i=1}^{n} (y_i - \bar{y})^2.$$
- $$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}).$$

Least Squares: A Mathematical Solution

- Our first estimating method about β_0 and β_1 make no. statistical assumption out the observations (x_i, y_i).
- Let

$$(x_1, y_1), \ldots, (x_n, y_n)$$

be n observation pairs, each of which consists of a measurement of X and a measurement of Y. In the Advertising example, this data set consists of the TV advertising budget and product sales in $n = 200$ different markets.

- Our goal: find an intercept $\widehat{\beta}_0$ and a slope $\widehat{\beta}_1$ such that the resulting line is as close as possible to the $n = 200$ data points.

Least Squares

- There are a number of ways of measuring closeness.
- However, by far the most common approach involves minimizing the least squares criterion.
- Let $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$ be the prediction for Y based on the ith value of X. Then $e_i = y_i - \widehat{y}_i$ represents the ith residual—this is the difference between the ith observed response value and the ith response value that is predicted by our linear model.
- The residual sum of squares (RSS):

$$\text{RSS}(\beta_0, \beta_1) = \sum_{i=1}^{n} e_i^2.$$

- LSEs of $\beta = (\beta_0, \beta_1)$ is given by

$$\widehat{\beta} = \mathit{argmin}_{\beta} \mathit{RSS}(\beta).$$

Least Squares

- The least squares approach chooses $\widehat{\beta}_0$ and $\widehat{\beta}_1$ to minimize the $RSS(\beta)$.
- $$\frac{\partial RSS(\beta)}{\partial \beta_0} = -2\sum_{i=1} n(y_i - (\beta_0 + \beta_1 x_i)),$$
$$\frac{\partial RSS(\beta)}{\partial \beta_1} = -2\sum_{i=1} n x_i(y_i - (\beta_0 + \beta_1 x_i).$$
- The minimizers are:
$$\widehat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}},$$
$$\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x},$$
where $\bar{y} = \sum_{i=1}^n y_i/n$ and $\bar{x} = \sum_{i=1}^n x_i/n$.

Least Squares

- The RSS is only one of many reasonable ways of measuring the distance from the line $c + dx$ to the data points.
- For example, rather than using vertical distance we could use horizontal distance. This is equivalent to graphing the y variable on the horizontal axis and the x variable on the vertical axis. Using the above result, we find the least squares line is $\widehat{x} = \tilde{\beta}_0 + \tilde{\beta}_1 y$, where $\tilde{\beta}_1 = \frac{S_{xy}}{S_{yy}}$ and $\tilde{\beta}_0 = \bar{x} - \tilde{\beta}_1 \bar{y}$. Reexpressing the line so that y is a function of x, we obtain $\widehat{y} = -(\tilde{\beta}_0/\tilde{\beta}_1) + (1/\tilde{\beta}_1)x$.

Least Squares

- Usually the line obtained by considering horizontal distance is different from the line obtained by considering vertical distance.
- The Least squares method should be considered as a method of fitting a line to a set of data, not as a method of statistical inference, as there is no basis for constructing confidence intervals and testing hypotheses.

Best Linear Unbiased Estimators: A Statistical Solution

- Here we show that the LE estimators are optimal in the class of linear unbiased estimators under a fairly general statistical model.
- Our model is:

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

where $\epsilon_1, \ldots, \epsilon_n$, called the random errors, are uncorrelated random variables with $E(\epsilon_i) = 0$ and $Var(\epsilon_i) = \sigma^2$.

Best Linear Unbiased Estimators: A Statistical Solution

- To derive estimators for β_1 in a linear model with an intercept, we restrict attention to the class of linear estimators, $\sum_{i=1} d_i Y_i$,
- In order for such estimators to be unbiased, we require that $\sum_{i=1}^n d_i = 0, \sum_{i=1}^n d_i x_i = 1$, because $\beta_1 = \beta_0(\sum_{i=1}^n d_i) + \beta_1 \sum_{i=1}^n d_i x_i$.
- BLUE=Best Linear Unbiased Estimator.
- Note that $Var(\sum_{i=1}^n d_i Y_i) = \sigma^2 \sum_{i=1}^n d_i^2$.
- To find the BLUE, we need to minimize

$$(1) \sum_{i=1}^n d_i^2$$

with the constraint that $\sum_{i=1}^n d_i = 0$ and $\sum_{i=1}^n d_i x_i = 1$.

Lemma 11.2.7

Let $(v_1, \ldots, v_k)$ be constants and let $(c_1, \ldots, c_k)$ be positive constants. Then,

$$max_{a_1,\ldots,a_k,\sum_i a_i=0}\{\frac{(\sum_{i=1}^k a_i v_i)^2}{\sum_{i=1}^k a_i^2/c_i}\} = \sum_{i=1}^k c_i(v_i - \bar{v}_c)^2,$$

where $\bar{v}_c = \sum_{i=1}^k c_i v_i / \sum_{i=1}^k c_i$. The maximum is obtained at any a of the form $a_i = Kc_i(v_i - \bar{v}_c)$, where K is a constant.

Solution

- To apply Lemma 11.2.7 to our case, we let $k = n$, $v_i = x_i$, $c_i = 1$, and $a_i = d_i$, which implies that $\bar{v}_c = \bar{x}$.
- If $d_i = Kc_i(v_i - \bar{v}_c) = K(x_i - \bar{x})$, $i = 1, \ldots, n$, by Lemma 11.2.7, $d_1, \ldots, d_n$ maximize

$$(2) \frac{(\sum_{i=1}^n d_i x_i)^2}{\sum_{i=1}^n d_i^2}$$

  among all $d_1, \ldots, d_n$ with $\sum_{i=1}^n d_i = 0$.
- Notice that $\{(d_1, \ldots, d_n) : \sum_i d_i = 0, \sum_i d_i x_i = 1\} \subset \{(d_1, \ldots, d_n) : \sum_i d_i = 0\}$.
- Hence, $d_1, \ldots, d_n$ that minimizes (1) also maximizes (2). (Since the set over which the maximum is taken is smaller, the maximum cannot be larger)

Solution

- Hence, $\sum_i d_i x_i = \sum_i K(x_i - \bar{x})x_i = KS_{xx}$.
- Hence, if we take $K = 1/S_{xx}$, then $\sum_i d_i x_i = 1$.
- Therefore,

$$d_i = \frac{(x_i - \bar{x})}{S_{xx}},$$

  which satisfies both constraints and also produces the maximum.
- Note that for all $d_i's$ that satisfies that $\sum_i d_i x_i = 1$ and $\sum_{i=1}^n d_i = 0$,

$$\frac{(\sum_{i=1}^n d_i x_i)^2}{\sum_{i=1}^n d_i^2} = \frac{1}{\sum_{i=1}^n d_i^2}$$

Solution

- The linear unbiased estimator for β_1 defined by those d_i, namely,

$$\widehat{\beta}_1 = \sum_{i=1}^{n} \frac{x_i - \bar{x}}{S_{xx}} y_i = \frac{S_{xy}}{S_{xx}}.$$

Geometric Description of BLUE

Geometric Description of BLUE

- Two planes represent the vectors (d_1, d_2, d_3) that satisfy the two linear constraints, and the line where the two planes intersect consists of the vectors (d_1, d_2, d_3) that satisfy both equalities.
- For any points on the line, $\sum_{i=1}^{n} d_i^2$ is the square of the distance from the points to the origin 0.
- The sphere in the figure is the smallest sphere that intersects the line, and point of intersection is the point (d_1, d_2, d_3) that defines the BLUE of β_1.
- $$Var(\widehat{\beta}_1) = \sigma^2 / S_{xx}$$
- We have shown that the LS estimator for β_1 is the BLUE of β_1. Similarly, we can show that the LE estimator for β_0 is also the BLUE of β_0 (Homework-Exercise 11.27).
- The fact that LS estimators are BLUEs in all linear models, is called Gauss-Markov Theorem.

Experimental Design

- Since $x_1, \ldots, x_n$ are values chosen by the experimenter, they can be chosen to make S_{xx} large and the variance of the estimator small.
- An experimenter can design the experiment to make the estimator more precise.
- Suppose all the $x_1, \ldots, x_n$ belong to an interval $[e, f]$. If n is even, the choice of $x_1, \ldots, x_n$ that makes S_{xx} as large as possible is to make the half of the x_is equal to e and half equal to f (Homework-Exercise 11.26).
- This two-point design is optimal in the sense that gives the most precise estimate of the slope β_1.
- If the linear model does not hold, this is not optimal

Advertising Data Example

- The least squares fit for the regression of sales onto TV is shown.
- The fit is found by minimizing the sum of squared errors.
- Each grey line segment: an error, and the fit makes a compromise by averaging their squares.
- A linear fit captures the essence of the relationship

Assessing the Accuracy of the Coefficient Estimates

- The population relationship:

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

  where the error term, ϵ is a catch-all for what we miss with this simple model: the true relationship is probably not linear, there may be other variables that cause variation in Y , and there may be measurement error.
- We typically assume that the error term is independent of X.

Conditional normal model

- To analyze the theoretical properties of the estimators, we assume that X is fixed, the conditional normal model.
- The observed data: $(x_1, y_1), \ldots, (x_n, y_n)$. The values of the predictor variable, $x_1, \ldots, x_n$ are considered to be fixed. The values of the response variable, $y_1, \ldots, y_n$ are observed values of random variables, $Y_1, \ldots, Y_n$ with the following distribution:

$$Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2).$$

  where Y_is are independent.
- Hence,

$$E(Y_i) = \beta_0 + \beta_1 x_i.$$

Bivariate normal model

- Sometimes, $x_1, \ldots, x_n$ are also observed values of random variables, $X_1, \ldots, X_n$. In this case, we need to consider models for both X and Y.
- Bivariate normal model:

$$(X_i, Y_i) \sim N_2((\mu_X, \mu_Y), \Sigma),$$

where

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \rho \\ \rho & \sigma_Y^2 \end{pmatrix}$$

- The conditional distribution of Y, given $X = x$:

$$E(Y \mid X = x) = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X) = [\mu_Y - \rho\frac{\sigma_Y}{\sigma_X}\mu_X] + [\rho\frac{\sigma_Y}{\sigma_X}]x.$$

Bivariate normal model

- It implies the following linear model:

$$E(Y \mid X = x) = \beta_0 + \beta_1 x$$

where $\beta_1 = \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X)$ and $\beta_0 = \mu_Y - \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X)$ and

$$Var(Y \mid X = x) = \sigma_Y^2(1 - \rho^2)$$

- For the bivariate normal model, the linear regression analysis is almost always carried out using the conditional distribution of $Y_1, \ldots, Y_n$ given $X_1 = x_1, \ldots, X_n = x_n$. Rather than the unconditional distribution of $(Y_1, X_1), \ldots, (Y_n, X_n)$.

Estimation and Testing with Normal Errors

- Consider the following model: $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, where $\epsilon_1, \ldots, \epsilon_n$ are iid $N(0, \sigma^2)$.
- The log-likelihood function:
  $l(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2} log(2\pi) - \frac{n}{2} log\sigma^2 - \frac{\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}$.
- Maximizing the log-likelihood is. equivalent to minimize $\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2$ which is equivalent to the LSE estimators,
$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}$$
- The MLE of σ^2:
$$\widehat{\sigma}^2 = \frac{1}{n}\sum_i (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i)^2$$

Estimation and Testing with Normal Errors

- We define the residuals from the regression: $\widehat{\epsilon}_i = Y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i$. Then, $\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n \widehat{\epsilon}_i^2 = \frac{1}{n} RSS$
- Since $E(Y_i) = \beta_0 + \beta_1 x_i$,

$$E(\widehat{\beta}_1) = \frac{E(S_{xy})}{S_{xx}} = \frac{\sum_{i=1}^n (x_i - \bar{x}) E(Y_i - \bar{Y})}{S_{xx}}.$$

- Note that $E(Y_i - \bar{Y}) = (x_i - \bar{x})\beta_1$. Hence, $E(\widehat{\beta}_1) = \beta_1$. Similarly, since $\widehat{\beta}_0 = \bar{Y} - \widehat{\beta}_1 \bar{x}$, we can show that $E(\widehat{\beta}_0) = \beta_0$.
- Hence, $E(\widehat{\epsilon}_i) = 0$.

Estimation and Testing with Normal Errors

- Therefore, (homework):
  $Var(\widehat{\epsilon_i}) = Var(Y_i) + Var(\widehat{\beta}_0) + x_i^2 Var(\widehat{\beta}_1) - 2Cov(Y_i, \widehat{\beta}_0) - 2x_i Cov(Y_i, \widehat{\beta}_1) + 2x_i Cov(\widehat{\beta}_0, \widehat{\beta}_1)$.
- We can show that $Cov(Y_i, \widehat{\beta}_0) = \sigma^2(\frac{1}{n} + \frac{(x_i - \bar{x})\bar{x}}{S_{xx}})$ and $Cov(Y_i, \widehat{\beta}_1) = \sigma^2 \frac{x_i - \bar{x}}{S_{xx}}$.

$$Var(\widehat{\epsilon_i}) = E(\widehat{\epsilon_i^2}) =$$

$$(\frac{n-2}{n} + \frac{1}{S_{xx}}[\frac{1}{n}\sum_{j=1}^{n} x_j^2 + x_i^2 - 2(x_i - \bar{x})^2 - 2x_i\bar{x})])\sigma^2$$

Estimation and Testing with Normal Errors

- $$E(\widehat{\sigma}^2) = \frac{n-2}{n}\sigma^2$$
- An unbiased estimator for σ^2:

$$S^2 = \frac{n}{n-2}\widehat{\sigma}^2$$

Estimation and Testing with Normal Errors

- To develop estimation and testing procedures, based on these estimators, we need to know their sampling distributions.
- Theorem 11.3.3 (Caella & Berger). Under the conditional normal regression model (11.3.22), the sampling distributions of the estimators $\widehat{\beta}_0$, $\widehat{\beta}_1$, and S^2 are

$$\widehat{\beta}_0 \sim N(\beta_0, \frac{\sigma^2}{nS_{xx}} \sum_{i=1}^{n} x_i^2), \widehat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{S_{xx}}),$$

  with $Cov(\widehat{\beta}_0, \widehat{\beta}_1) = -\frac{\sigma^2 \bar{x}}{S_{xx}}$.
- Furthermore, $(\widehat{\beta}_0, \widehat{\beta}_1)$ and S^2 are independent and

$$\frac{(n-2)S^2}{\sigma^2} \sim \chi^2_{n-2}.$$

Proof of Theorem 11.3.3

Proof:

- Denote $X = \begin{pmatrix} 1 & x_1 \\ \dots & \dots \\ 1 & x_n \end{pmatrix}$, $\epsilon = (\epsilon_1, \dots, \epsilon_n)^T$, and $\hat{\epsilon} = (\hat{\epsilon_1}, \dots, \hat{\epsilon_n})^T$.
- It's easy to show that $\begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = (X^T X)^{-1} X^T Y = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + (X^T X)^{-1} X^T \epsilon$. As a result, $\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} \sim N(\beta, (X^T X)^{-1} \sigma^2)$.

Proof of Theorem 11.3.3, continued

- $n\hat{\sigma}^2 = \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \hat{\epsilon}^T\hat{\epsilon}$.
- Notice that $\hat{\beta} = (X^TX)^{-1}X^TY$ and

$$\hat{\epsilon} = Y - X\hat{\beta}$$

$$= Y - X(X^TX)^{-1}X^TY = (I - X(X^TX)^{-1}X^T)Y.$$

We have

$$Cov(\hat{\beta}, \hat{\epsilon}) = (X^TX)^{-1}X^T Cov(Y, Y)(I - X(X^TX)^{-1}X^T)^T$$

$$= (X^TX)^{-1}X^T(\sigma^2 I)(I - X(X^TX)^{-1}X^T)^T$$

$$= (X^TX)^{-1}X^T\sigma^2 - (X^TX)^{-1}X^TX(X^TX)^{-1}X^T\sigma^2 = 0.$$

This proves that S^2 and $\hat{\beta}$ are independent.

Proof of Theorem 11.3.3, continued

- Moreover,
  $\hat{\epsilon} = (I - X(X^T X)^{-1} X^T)(X\beta + \epsilon) = (I - X(X^T X)^{-1} X^T)\epsilon.$
- We have

$$\sum_{i=1}^{n} \hat{\epsilon}_i^2 = \epsilon^T (I - X(X^T X)^{-1} X^T)^T (I - X(X^T X)^{-1} X^T)\epsilon$$

$$= \epsilon^T (I - X(X^T X)^{-1} X^T)\epsilon.$$

- We can show that
  $(I - X^T(X^T X)^{-1} X)(I - X^T(X^T X)^{-1} X) =$
  $I - X^T(X^T X)^{-1} X$, which means it is an idempotent matrix. Any of its eigenvalues is either 0 or 1.

Proof of Theorem 11.3.3, continued

- Also we have

$$trace(I - X(X^TX)^{-1}X^T) = n - tr(X(X^TX)^{-1}X^T)$$

$$= n - tr((X^TX)^{-1}X^TX) = n - 2,$$

which means $I - X(X^TX)^{-1}X^T$ is similar to a diagonal matrix with n-2 diagonal elements of value 1 and 2 diagonal elements of value 0.

- Suppose $I - X(X^TX)^{-1}X^T = G^T diag\{1, 1, \ldots, 1, 0, 0\}G$. We have $\epsilon^T(I - X(X^TX)^{-1}X^T)\epsilon/\sigma^2 = (\epsilon^T G^T/\sigma)diag\{1, 1, \ldots, 1, 0, 0\}(G\epsilon/\sigma)$.
where $G\epsilon/\sigma$ is a vector of independent standard normal variables. As a result, we showed that $\sum_{i=1}^{n} \hat{\epsilon}_i^2/\sigma^2 \sim \chi^2_{n-2}$.

Assessing the Accuracy of the Coefficient Estimates, continued

- We would like to consider how close $\widehat{\beta}_0$ and $\widehat{\beta}_1$ are to the true values β_0 and β_1.
- We can show the the following standard errors associated with $\widehat{\beta}_0$ and $\widehat{\beta}_1$:

$$SE(\widehat{\beta}_0)^2 = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right),$$

$$SE(\widehat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2},$$

where $\sigma^2 = Var(\epsilon)$.

Assessing the Accuracy of the Coefficient Estimates, continued

- For linear regression, we can use a t-statistic with $n-2$ degrees of freedom of the form

$$t = \frac{\widehat{\beta}_1}{SE(\widehat{\beta}_1)}$$

  to test whether $H_0 : \beta_1 = 0$ versus $H_a : \beta_1 \neq 0$.
- More generally, we can use an F-test for the whole model, or subset of parameters

Assessing the Accuracy of the Model

- We like to assess how well the model fits the data.
- The quality of a linear regression fit is typically assessed using two related quantities: the residual standard error (RSE) and the R^2 statistic.
- Note $y_i = \hat{y}_i + (y_i - \hat{y}) = \hat{y}_i + e_i$ and $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i)$.
- Hence, it can be shown (homework) that

$$SST = SSR + SSE,$$

true for both simple and multiple regression models, where

$$SSR = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2,$$

and

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

Residual Standard Error (RSE)

- Note that the linear regression model that associated with each observation is an error term ϵ.
- Due to the presence of these error terms, even if we knew the true regression line (i.e. even if β_0 and β_1 were known), we would not be able to perfectly predict Y from X.
- The RSE is an estimate of the standard deviation of ϵ.

Residual Standard Error (RSE)

- Roughly speaking, RSE is the average amount that the response will deviate from the true regression line. It is computed using the formula

$$RSE = \sqrt{\frac{1}{n-2}RSS},$$

where

$$RSS = \sum_{i=1}^{n}(y_i - \widehat{y}_i)^2.$$

- The RSE is considered a measure of the lack of fit of the model to the data.

R^2 Statistics

- The RSE provides an absolute measure of lack of fit of the linear model to the data.
- But since it is measured in the units of Y , it is not always clear what constitutes a good RSE.
- The R^2 statistic provides an alternative measure of fit. It takes the form of a proportion - the proportion of variance explained - and so it always takes on a value between 0 and 1, and is independent of the scale of Y.

R^2 Statistics

- The R^2 statistic provides an alternative measure of fit. It takes the form of a proportion - the proportion of variance explained - and so it always takes on a value between 0 and 1, and is independent of the scale of Y.
- $$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

  where $TSS = \sum_{i=1}^{n}(y_i - \bar{y})^2$ is the total sum of squares. TSS measures the total variance in the response Y , and can be thought of as the amount of variability inherent in the response before the regression is performed.

R^2 Statistics

- An R^2 statistic that is close to 1 indicates that a large proportion of the variability in the response has been explained by the regression.
- A number near 0 indicates that the regression did not explain much of the variability in the response; this might occur because the linear model is wrong, or the inherent error σ^2 is high, or both.

Relationship between R^2 and Correlation

- The R^2 statistic is a measure of the linear relationship between X and Y .
- Another way of interpreting the coefficient of determination R^2 is to look at it as the sample Squared Pearson Correlation Coefficient between the observed n paired values $(y_i, \hat{y}_i)$'s or n paris (y_i, x_i)'s.
- Let us define the following correlations:

$$r_1 = \widehat{Corr}(Y, \widehat{Y}) = \frac{S_{y\widehat{y}}}{\sqrt{S_{yy} S_{\widehat{y}\widehat{y}}}}$$

$$r_2 = \widehat{Corr}(Y, X) = \frac{S_{xy}}{\sqrt{S_{xx}}\sqrt{S_{yy}}}.$$

- Theorem (homework): Under the simple linear model, it can be shown that

$$R^2 = r_1^2 = r_2^2.$$

Proof

- $$y_i = \hat{y}_i + (y_i - \hat{y}) = \hat{y}_i + e_i$$
- $$y_i - \bar{y} = (\hat{y}_i - \bar{y}) + (y_i - \hat{y}_i).$$
- Hence,
$$SST = SSR + SSE,$$
true for both simple and multiple regression models, where $SSR = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$, and $SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$.
- After some algebra calculation,
$$R^2 = r_1^2 = r_2^2$$

Estimation and Prediction

- Estimation for

$$E(Y \mid X = x_)) = \beta_0 + \beta_1 x :$$

$\widehat{\beta}_0 + \widehat{\beta}_1 x_0$.

- We have the following result:

$$\widehat{\beta}_0 + \widehat{\beta}_1 x_0 \sim N(\beta_0 + \beta_1 x_0, \sigma^2(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})),$$

and

$$\frac{\widehat{\beta}_0 + \widehat{\beta}_1 x_0 - (\beta_0 + \beta_1 x_0)}{S\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}} \sim t_{n-1}$$

Estimation and Prediction

Proof:

- $Var(\widehat{\beta}_0 + \widehat{\beta}x_0) = Var(\widehat{\beta}_0) + Var(\widehat{\beta}_1)x_0^2 + 2x_0 Cov(\widehat{\beta}_0, \widehat{\beta}_1) = \frac{\sigma^2}{nS_{xx}} \sum_{i=1}^n x_i^2 + \frac{\sigma^2 x_0^2}{S_{xx}} - \frac{2\sigma^2 x_0 \bar{x}}{S_{xx}} = \frac{\sigma^2}{S_{xx}}(\frac{1}{n}\sum_{i=1}^n x_i^2 - \bar{x}^2 + \bar{x}^2 - 2x_0\bar{x} + x_0^2) = \frac{\sigma^2}{S_{xx}}(\frac{1}{n}[\sum_{i=1}^n x_i^2 - \frac{1}{n}(\sum_{i=1}^n x_i)^2] + (x_0 - \bar{x})^2)$. Because $\sum x_i^2 - \frac{1}{n}(\sum_i x_i)^2 = S_{xx}$,

$$Var(\widehat{\beta}_0 + \widehat{\beta}x_0) = \sigma^2(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})$$

- Since $\widehat{\beta}_0$ and $\widehat{\beta}_1$ are both linear functions of $Y_1, \ldots, Y_n$, so is $\widehat{\alpha} + \widehat{\beta}x_0$. Thus. $\widehat{\beta}_0 + \widehat{\beta}x_0$ has a normal distribution.
- Since $(\widehat{\beta}_0, \widehat{\beta}_1)$ and S^2 are independent, S^2 is also independent of $\widehat{\beta}_0 + \widehat{\beta}_1 x_0$ and $\frac{\widehat{\beta}_0 + \widehat{\beta}_1 x_0 - (\beta_0 + \beta_1 x_0)}{s\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}} \sim t_{n-1}$

Estimation and Prediction

- The above pivot can be inverted to give the $100(1-\alpha)\%$ confidence interval for $\beta_0 + \beta_1 x_0$:

$$\widehat{\beta}_0 + \widehat{\beta}_1 x_0 - t_{n-2,1-\alpha/2} S \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

$$\leq \beta_0 + \beta_1 x_0 \leq \widehat{\beta}_0 + \widehat{\beta}_1 x_0 + t_{n-2,1-\alpha/2} S \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

- The length of the confidence interval for $\beta_0 + \beta_1 x_0$ depends the values of $x_1, \ldots, x_n$ and is minimized at $x_0 = \bar{x}$.

Prediction

- A $100(1-\alpha)\%$ prediction interval for an unobserved random variable Y based on the observed data X is a random interval $[L(X), U(X)]$ with the property that

$$P_\theta(L(X) \leq Y \leq U(X)) \geq 1-\alpha$$

  for all θ.
- Since a random variable is more variable than a parameter, we expect a prediction interval to be wider than a parameter.
- We assume that a new observation Y_0 to be taken at $x = x_0$ has a $N(\beta_0 + \beta_1 x_0, \sigma^2)$, independent of the previous data, $(x_1, Y_1), \ldots, (x_n, Y_n)$.

Prediction

- Then, we find that $Y_0 - (\widehat{\beta}_0 + \widehat{\beta}_1 x_0)$ has a normal distribution with mean $E(Y_0 - (\widehat{\beta}_0 + \widehat{\beta}_1 x_0)) = 0$ and variance

$$Var(Y_0 - (\widehat{\beta}_0 + \widehat{\beta}_1 x_0)) = Var(Y) + Var(\widehat{\beta}_0 + \widehat{\beta}_1 x_0) =$$

$$\sigma^2 + \sigma^2(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})$$

- The $100(1-\alpha)\%$ prediction interval for Y_0 is:

$$\widehat{\beta}_0 + \widehat{\beta}_1 x_0 - t_{n-2,1-\alpha/2} S \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

$$\leq Y_0 \leq \widehat{\beta}_0 + \widehat{\beta}_1 x_0 + t_{n-2,1-\alpha/2} S \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

Simultaneous Estimation and Confidence Bands

- In the above discussion, we have provided formulas for confidence intervals for one single value of x_0.
- Sometimes We may be interested in confidence intervals for the Y population means at a number of x_0 values, $E(Y \mid x_{0i}), i = 1, \ldots, m$.
- Using the Bonferroni Inequality, we can obtain that the probability is at least $1 - \alpha$ that

$$\widehat{\beta}_0 + \widehat{\beta}_1 x_{0i} - t_{n-2,1-\alpha/(2m)} S \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

$$\leq \beta_0 + \beta_1 x_{0i} \leq \widehat{\beta}_0 + \widehat{\beta}_1 x_{0i} + t_{n-2,1-\alpha/(2m)} S \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

simultaneously for all $i = 1, \ldots, m$.

Simultaneous Estimation and Confidence Bands

Theorem 11.3.6:

- More generally, under the conditional normal regression model (11.3.22), the probability is at least $1-\alpha$ that

$$\widehat{\beta}_0 + \widehat{\beta}_1 x_0 - M_\alpha S \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

$$\leq \beta_0 + \beta_1 x_0 \leq \widehat{\beta}_0 + \widehat{\beta}_1 x_0 + M_\alpha S \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$$

  simultaneously for all x_0, where $M_\alpha = \sqrt{2F_{2,n-2,1-\alpha}}$
- Lemma: If a, b, c, and d are constant, with $c > 0$ and $d > 0$, we have the following result:

$$max_t \frac{(a+bt)^2}{c+dt^2} = \frac{a^2}{c} + \frac{b^2}{d}$$

- Denote $\alpha = \beta_0$ and $\beta = \beta_1$.

Proof

- If we rearrange terms, we can get that the conclusion is true if we can find a constant M_α that satisfies

$$P\left(\frac{((\widehat{\beta}_0 + \widehat{\beta}_1 x_0) - (\beta_0 + \beta_1 x_0))^2}{S^2(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})} \leq M_\alpha^2 \text{ for all } x_0\right) = 1 - \alpha$$

- Or, equivalently

$$P\left(max_{x_0} \frac{((\widehat{\beta}_0 + \widehat{\beta}_1 x_0) - (\beta_0 + \beta_1 x_0))^2}{S^2(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})} \leq M_\alpha^2\right) = 1 - \alpha$$

- We can write that $\widehat{\beta}_0 + \widehat{\beta}_1 x = \bar{Y} + \widehat{\beta}_1(x - \bar{x})$, $\beta_0 + \beta_1 x = \mu_{\bar{Y}} + \beta_1(x - \bar{x})$, where $\mu_{\bar{Y}} = E(\bar{Y}) = \beta_0 + \beta_1 \bar{x}$.

Proof, continued

- If we let $t = x = \bar{x}$, we have that

$$\frac{((\widehat{\beta}_0 + \widehat{\beta}_1 x_0) - (\beta_0 + \beta_1 x_0))^2}{S^2(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})} = \frac{((\bar{Y} - \mu_{\bar{Y}}) + (\widehat{\beta}_1 - \beta_1)t)^2}{S^2(\frac{1}{n} + \frac{t^2}{S_{xx}})}$$

- Or, equivalently we like to find M_α such that

$$P(max_t \frac{((\bar{Y} - \mu_{\bar{Y}}) + (\widehat{\beta}_1 - \beta_1)t)^2}{S^2(\frac{1}{n} + \frac{t^2}{S_{xx}})} \leq M_\alpha^2) = 1 - \alpha$$

- We can write that $\widehat{\beta}_0 + \widehat{\beta}_1 x = \bar{Y} + \widehat{\beta}_1(x - \bar{x})$, $\beta_0 + \beta_1 x = \mu_{\bar{Y}} + \beta_1(x - \bar{x})$, where $\mu_{\bar{Y}} = E(\bar{Y}) = \beta_0 + \beta_1 \bar{x}$.

Proof, continued

- Using the result of the above lemma, we obtain that

$$max_t \frac{((\bar{Y} - \mu_{\bar{Y}}) + (\widehat{\beta}_1 - \beta_1)t)^2}{S^2(\frac{1}{n} + \frac{t^2}{S_{xx}})} = \frac{\frac{(\bar{Y} - \mu_{\bar{Y}})^2}{\sigma^2/n} + \frac{(\widehat{\beta}_1 - \beta_1)^2}{\sigma^2/S_{xx}}}{S^2/\sigma^2}$$

- From Theorem 11.3.3. of Casella and Berger's book, we can obtain that the last expression is the ratio of two independent chi squared random variables, the denominator being divided by its degrees of freedom. The numerator is the sum of two independent random variables, each of which has a χ^2_1 distribution. Therefore, the numerator is distributed as χ^2_2.
- Hence, the distribution of the quotient has the following result:

$$\frac{\frac{(\bar{Y} - \mu_{\bar{Y}})^2}{\sigma^2/n} + \frac{(\widehat{\beta}_1 - \beta_1)^2}{\sigma^2/S_{xx}}}{S^2/\sigma^2} \sim 2F_{2,n-2},$$

Proof, continued

- Hence, we have the following result:

$$P(max_t \frac{((\bar{Y} - \mu_{\bar{Y}}) + (\widehat{\beta}_1 - \beta_1)t)^2}{S^2(\frac{1}{n} + \frac{t^2}{S_{xx}})} \leq M_\alpha^2) = 1 - \alpha,$$

where $M_\alpha = \sqrt{2F_{2,n-2,1-\alpha}}$.